THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 13: **Statistics** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In a distribution, $\sum f_i d_i = -60$, $\sum f_i = 30$ and the assumed mean is 50. The mean is: 1

(a) -2 **(b)** 50 **(c)** 48 **(d)** 52

2. For the data with classes 80–90, 90–100, 100–110, 110–120, 120–130, 130–140 with frequencies 10, 3, 3, 7, 20, 7 respectively, The mean of the data is: 1

(a) 114 **(b)** 950 **(c)** 113 **(d)** 115

3. For the data with classes 40–50, 50–60, 60–70, 70–80, 80–90, 90–100 with frequencies 8, 14, 22, 20, 17, 19 respectively, The median of the data is: 1

(a) 70 **(b)** 73 **(c)** 80 **(d)** 73.1

4. For the data with classes 60–70, 70–80, 80–90, 90–100, 100–110, 110–120, 120–130 with frequencies 3, 16, 5, 17, 20, 21, 18 respectively, The mode of the data is: 1

(a) 120 **(b)** 102 **(c)** 112.5 **(d)** 110

5. The mean of a distribution with classes 40–50, 50–60, 60–70, 70–80, 80–90, 90–100, 100–110 and frequencies 17, 21, 9, 15, p, 3, 20 respectively is 72.9. The value of p is: 1

(a) 17 **(b)** 20 **(c)** 14 **(d)** 15

6. The median of a distribution with classes 40–50, 50–60, 60–70, 70–80, 80–90, 90–100, 100–110 and frequencies 17, x, 20, y, 18, 6, 15 respectively, and total frequency 100, is 67.5. The value of x is: 1

(a) 18 **(b)** 17 **(c)** 19 **(d)** 8

7. The median of a distribution with classes 40–50, 50–60, 60–70, 70–80, 80–90, 90–100, 100–110 and frequencies 6, 14, x, 18, y, 18, 14 respectively, and total frequency 100, is 75. The value of y is: 1

(a) 11 **(b)** 9 **(c)** 20 **(d)** 22

8. If every observation of a grouped data is increased by 5 (every class shifted by 5), the mean: 1

(a) decreases by 5 **(b)** increases by 5 **(c)** becomes 5 times **(d)** remains the same

9. For the data with classes 20–30, 30–40, 40–50, 50–60, 60–70, 70–80, 80–90 with frequencies 11, 19, 7, 20, 13, 19, 11 respectively, The mean of the data is: 1

(a) 55 **(b)** 55.6 **(c)** 54.6 **(d)** 56.6

10. For the data with classes 0–10, 10–20, 20–30, 30–40, 40–50, 50–60 with frequencies 17, 9, 16, 20, 21, 17 respectively, The median of the data is: 1

(a) 30 **(b)** 40 **(c)** 32 **(d)** 34

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. The ages (in years) of patients admitted to a hospital in a month are given below. 2

Age (years)	40–50	50–60	60–70	70–80	80–90	90–100
Number of patients	5	7	4	20	11	3

Find the mode.

12. The weights (in kg) of students of a class are given below. 2

Weight (kg)	80–90	90–100	100–110	110–120	120–130	130–140
Number of students	10	3	3	7	20	7

Find the mean.

13. The weights (in kg) of students of a class are given below. 2

Weight (kg)	70–80	80–90	90–100	100–110	110–120	120–130	130–140
Number of students	14	11	6	20	14	18	17

Find the median.

14. The weights (in kg) of students of a class are given below. 2

Weight (kg)	60–70	70–80	80–90	90–100	100–110	110–120	120–130
Number of students	3	16	5	17	20	21	18

Find the mode.

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The daily electricity consumption (in units) of households in a colony is given below. 3

Units consumed	0–10	10–20	20–30	30–40	40–50	50–60
Number of households	17	9	16	20	21	17

Find the mean.

16. The monthly pocket money (in ₹) of students is given below. 3

Pocket money (₹)	20–30	30–40	40–50	50–60	60–70	70–80	80–90
Number of students	11	19	7	20	13	19	11

Find the median.

17. The daily electricity consumption (in units) of households in a colony is given below. 3

Units consumed	40–50	50–60	60–70	70–80	80–90	90–100	100–110
Number of households	17	21	9	15	15	3	20

Find the mean and the mode (in this order).

OR

The lifetimes (in hours) of electric bulbs tested in a laboratory are given below. 3

Lifetime (hours)	60–70	70–80	80–90	90–100	100–110	110–120	120–130
Number of bulbs	2	16	10	2	4	5	11

Find the mean.

18. The marks obtained by students in a test are given below. 3

Marks	0–10	10–20	20–30	30–40	40–50	50–60
Number of students	10	6	15	12	5	2

Find the median.

OR

The lifetimes (in hours) of electric bulbs tested in a laboratory are given below. 3

Lifetime (hours)	30–40	40–50	50–60	60–70	70–80	80–90
Number of bulbs	11	8	16	8	3	4

Find the mean and the mode (in this order).

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. The mean of a distribution with classes 20–30, 30–40, 40–50, 50–60, 60–70, 70–80, 80–90 and frequencies 11, 19, 7, p, 13, 19, 11 respectively is 55.6. Find the value of p. 5

OR

The median of a distribution with classes 60–70, 70–80, 80–90, 90–100, 100–110, 110–120, 120–130 and frequencies 10, 18, x, 10, y, 10, 12 respectively, and total frequency 100, is 94. Find the value of x. 5

20. The mean of a distribution with classes 0–10, 10–20, 20–30, 30–40, 40–50, 50–60 and frequencies 10, 6, p, 12, 5, 2 respectively is 25.4. Find the value of p. 5

OR

The median of a distribution with classes 20–30, 30–40, 40–50, 50–60, 60–70, 70–80, 80–90 and frequencies 20, 15, 3, x, 20, y, 17 respectively, and total frequency 100, is 64. Find the value of x. 5

END OF QUESTION PAPER — **Total: 40 Marks**